

Gas Pressure, the Particle Model and Diffusion — Paper D

HARD • 50 MINUTES • 30 MARKS

NAME

DATE

SCORE / 30

- Answer every question in the space provided. Show your working — method marks are awarded even when the final answer is wrong.
- A calculator is allowed, but every number on this paper is designed to work without one.
- Every numerical answer must carry a unit. An answer with no unit does not score the final mark.
- Explanations must be written in terms of particles. “The particles get bigger”, “the particles expand” and “the particles push harder because they are squashed” score nothing.
- The mark for each question is shown in square brackets on the right.
- No formula is given on this paper, and no value for atmospheric pressure.
- Several questions describe apparatus you may not have seen before — apply the ideas you already have.
- Where an explanation is worth three marks or more, the mark scheme is looking for collisions per unit area per second, and, wherever the temperature changes, for the force of each collision as well.

SECTION 1 – APPARATUS YOU MAY NOT HAVE MET

- 1.** A pressure cooker is a sealed pot with a locking lid and a valve in the top. Water is boiled inside it and the volume of the pot cannot change.

- (a)** Explain, in terms of particles, why the pressure inside rises as the pot is heated. [2]

- (b)** The valve opens and lets steam escape when the pressure reaches 200 kPa. [2]
Explain, in terms of particles, why this lowers the pressure inside.

- (c)** Explain why the lid must be locked shut. [1]

- 2.** A weather balloon is released at ground level, where atmospheric pressure is 100 kPa. It rises until the atmospheric pressure around it is only 25 kPa. Assume the temperature does not change.

(a) Explain, in terms of particles, why atmospheric pressure falls as you go higher. [2]

(b) At ground level the balloon has a volume of 2.0 m^3 . Calculate its volume at 25 kPa. [2]

(c) Suggest what eventually happens to the balloon, and why. [1]

SECTION 2 – REASONING IT OUT

- 3.** Two identical sealed containers are kept at the same temperature. Container X holds twice as many gas particles as container Y.

(a) State which container has the greater pressure, and by what factor. [1]

(b) Explain your answer in terms of particles. [2]

(c) Y is at 300 K. Calculate the temperature Y must be heated to for its pressure to match X. [1]

- 4.** A diffusion tube like the ammonia and hydrogen chloride one is set up, but this time the two gases have the same relative molecular mass, and the tube is kept at a steady temperature.

(a) Predict where the ring will form, and explain your prediction. [2]

(b) The tube is now stood vertically, with the denser gas at the top. Suggest one reason the result may no longer be a fair test of diffusion. [2]

5. A diver's air cylinder holds gas at a pressure of 20 000 000 Pa. The valve at the top has an area of 0.0004 m^2 , and atmospheric pressure outside it is 100 000 Pa.

(a) Calculate the force the compressed gas exerts on the inside of the valve. [2]

(b) Calculate the force the atmosphere exerts on the outside of the valve, and hence the resultant force. [2]

SECTION 3 – JUDGEMENT

6. A bag of crisps is sealed in a factory at sea level and driven up a mountain road. At the top it has swollen up tightly, although nobody has opened it and it has not been warmed. Explain what has happened, in terms of particles and pressure. [4]

- 7.** A student has written two answers, and both are wrong. [4]
- (i) "When you squash a gas, the particles are pushed closer together so they push harder on the walls. That is why the pressure goes up."
- (ii) "The can was crushed because the vacuum inside sucked the sides in."
- For each one, explain the mistake and write the correct version.

End of paper. Check every answer has a unit or a form the question actually asked for.